

JWST Synthesis Calculator – Multi-Instrument UQFF Cross-Reference

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2026-04-13

PAPER_1071: JWST Synthesis Calculator — Multi-Instrument UQFF Cross-Reference

Abstract

We compute UQFF predictions cross-referenced with JWST instrument capabilities. NIRCам (0.6-5 μm) maps to SCm phonon wavelength $\lambda_{\text{SCm}} = c / f_{\text{SCm}} = 240 \text{ } \mu\text{m}$ (MIRI-accessible). MIRI (5-28 μm) maps to the SCm axiom validation wavelength 25.4 μm ($|U_b/U_g| > 0.5$). NIRSpec (0.6-5.3 μm) resolves spectral features at $R = 2700$ for phonon absorption line searches. Predicted SCm absorption depth: $\delta F / F$ approx 10^{-6} at 25.4 μm .

1. Key Equations

- SCm axiom wavelength: 25.4 μm (MIRI); absorption depth: $\delta F / F \sim 10^{-6}$
- NIRCам + MIRI + NIRSpec cross-reference to SCm predictions

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class JWSTSynthesisCalculator.

References

- PAPER_1070

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[\text{SSq}]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance

Constant	Symbol	Value	Validation Domain
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: observational (JWST instruments)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = 0$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> observational (JWST instruments) -> phonon coupling -> UQFF prediction -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed